Name

Mathematics Extension 1 Year 12

Volumes of Rotation

Book 1

Version: 261028

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Contents

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Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

ME1-12-05 applies calculus to solve problems involving polynomials, further rates of change, areas and volumes and differential equations

Areas between curves and volumes of solids of revolution

- Calculate areas of regions between curves determined by functions in both real-life and abstract contexts
- Examine a solid of revolution whose boundary is formed by rotating an arc of a function about the x-axis or y-axis with and without graphing applications
- Calculate the volume of a solid of revolution formed by rotating a region in the plane about the x-axis or y-axis in both real-life and abstract contexts
- Calculate the volume of a solid of revolution formed by rotating the region between two curves about either the x-axis or y-axis in both real-life and abstract contexts

Area Between Two Curves

Review

☐ Find point of intersection of two curves and sketch both curves on the same axes.

a $y = x^2$ and $y = x + 2$

b $y = x^2$ and $y = x + 6$

c $y = x^3$ and $y = 4x$

d $y = (x - 1)^2$ and $y = (x + 1)^2$

e $y = x$ and $y = 2(x - 3)^2$

f $y = x(x - 4)^2$ and $y = x$

Area Under a Combination of Curves

Sometimes a region is bounded by different curves in different sections along the x-axis. Because no single curve forms the top boundary across the whole region, you must split the region at the point(s) of intersection and calculate each section separately.

1. Find the intersection points.
2. Set up separate definite integrals for each section.
3. Integrate and add the results.

Example Find area under a combination of curves

Find the area bounded by $y = x^2$, $y = (x-4)^2$ and the axis.

1. Intersection points

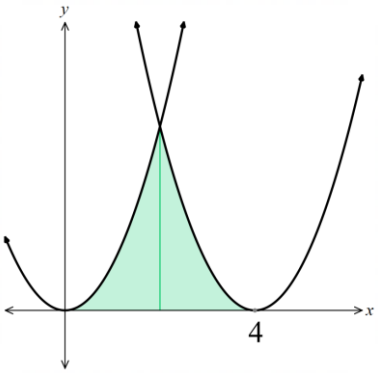
$$x^2 = (x-4)^2$$
$$x^2 = x^2 - 8x + 16$$
$$x = 2$$

2. Set up integrals

$$\int_0^2 x^2\,dx + \int_2^4 (x-4)^2\,dx$$

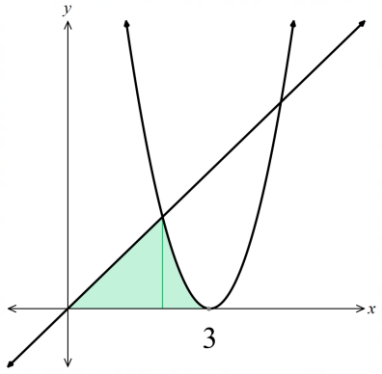
3. Integrate and add

$$= \left[\frac{x^3}{3}\right]_0^2 + \left[\frac{(x-4)^3}{3}\right]_2^4$$
$$= \left(\frac{8}{3} - 0\right) + \left(0 - \frac{-8}{3}\right)$$
$$= \frac{16}{3}\ \text{u}^2$$



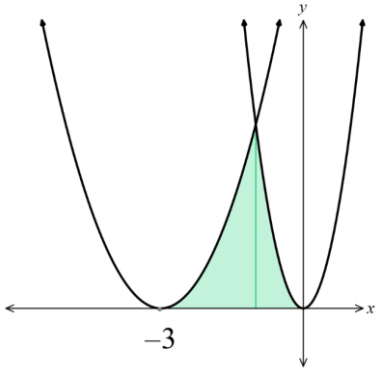
Guided Practice

a Find the area of the bounded by $y = x$, $y = 2(x-3)^2$, and the x-axis.



$\frac{8}{3}$

b Find the area of the bounded by $y = 4x^2$, $y = (x+3)^2$, and the x-axis.



4

Area Between Two Curves

When two curves intersect at two points, they enclose a region between them.
To find this area, integrate the **difference between the upper curve and the lower curve**.

1. Find the intersection points.
2. Identify the upper curve; the one with the larger y-values is the upper curve.
   A sketch is useful, you can also substitute x values into each function to check.
3. Set up the integral as $\int (\text{upper curve} - \text{lower curve})\, dx$.
   If the curves cross each other in the domain, split the region into subintervals.
   Simplify the integrand if possible.
4. Integrate and evaluate.

Example Find area between two curves

Find the area enclosed between $y = x^2$ and $y = x + 2$.

1. Intersection points

$x^2 = x + 2$

$x^2 - x - 2 = 0$

$(x - 2)(x + 1) = 0$

$x = -1, 2$

2. Identify upper curve

In the required region,

$y = x + 2$ is the upper curve

3. Set up integral

$$\int_{-1}^{2} \left((x + 2) - (x^2)\right) dx$$

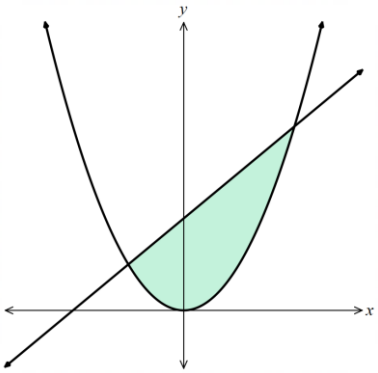
$$= \int_{-1}^{2} (x + 2 - x^2)\, dx$$

4. Integrate

$$= \left[\frac{x^2}{2} + 2x - \frac{x^3}{3}\right]_{-1}^{2}$$

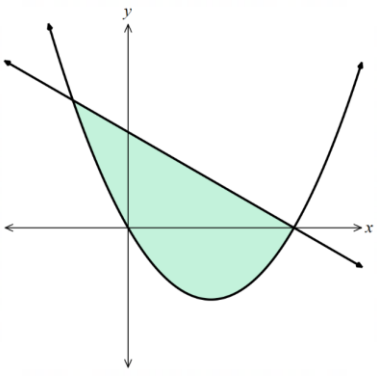
$$= \frac{10}{3} - \left(-\frac{7}{6}\right)$$

$$= 4.5 \text{ u}^2$$



Guided Practice

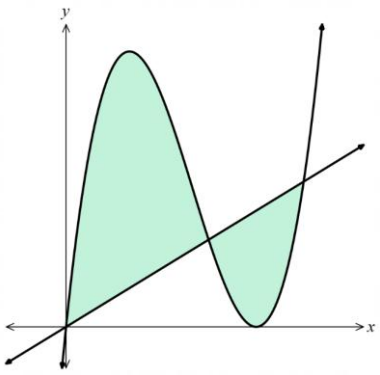
a Find the area enclosed between $y = 3 - x$ and $y = x^2 - 3x$.



$\frac{32}{3}$ u²

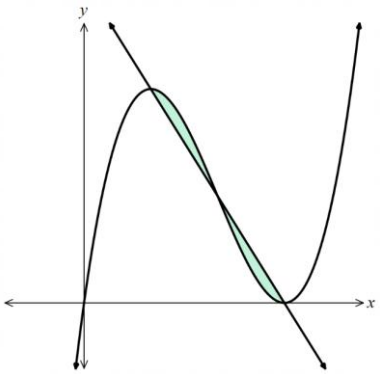
Guided Practice

b Find the area bounded between $y = x(x-4)^2$ and $y = x$



$\frac{253}{12}$ u²

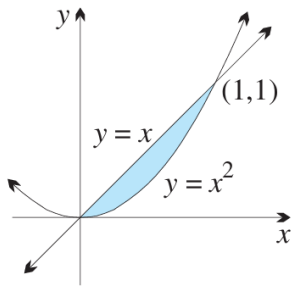
c Find the area bounded between $y = 6 - 2x$ and $y = x(x-3)^2$



$\frac{1}{2}$ u²

1 Find the area of the shaded region in each diagram below:

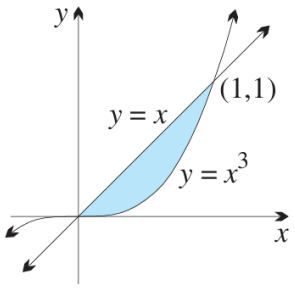
a



$A = \int_0^1 (x - x^2)\,dx$

$= \frac{1}{6}$ u²

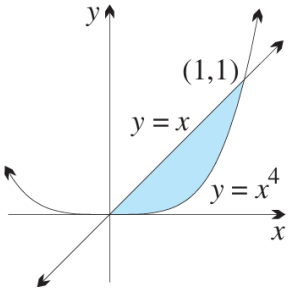
b



$A = \int_0^1 (x - x^3)\,dx$

$= \frac{1}{4}$ u²

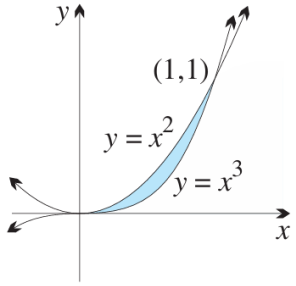
c



$A = \int_0^1 (x - x^4)\,dx$

$= \frac{3}{10}$ u²

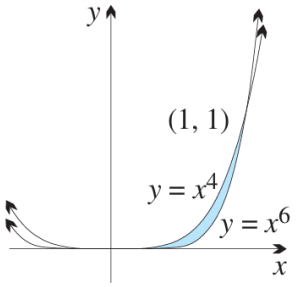
d



$A = \int_0^1 (x^2 - x^3)\,dx$

$= \frac{1}{12}$ u²

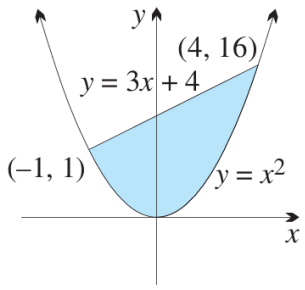
e



$A = \int_0^1 (x^4 - x^6)\,dx$

$= \frac{2}{35}$ u²

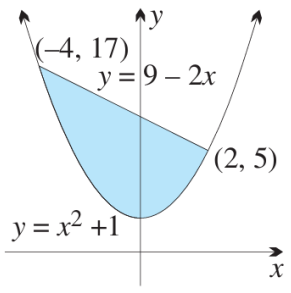
f



$A = \int_{-1}^4 (3x + 4 - x^2)\,dx$

$= \frac{125}{6}$ u²

g

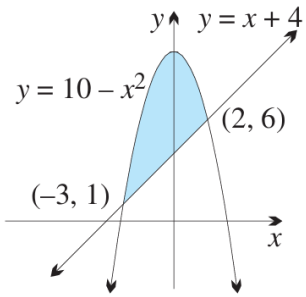


$A = \int_{-4}^2 (9 - 2x - x^2 - 1)\,dx$

$= \int_{-4}^2 (8 - 2x - x^2)\,dx$

$= 36$ u²

h



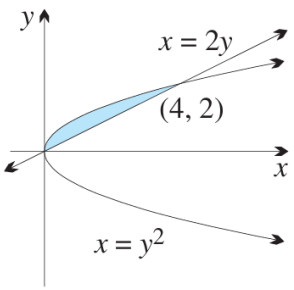
$A = \int_{-3}^2 (10 - x^2 - x - 4)\,dx$

$= \int_{-3}^2 (6 - x - x^2)\,dx$

$= \frac{125}{6}$ u²

2 By considering regions between the curves and the y-axis, find the area of the shaded region.

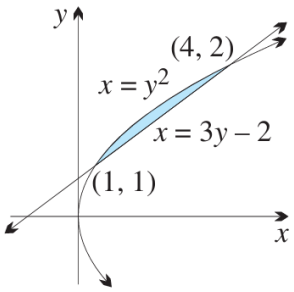
a



$A = \int_0^2 (2y - y^2)\,dy$

$= \frac{4}{3}$ u²

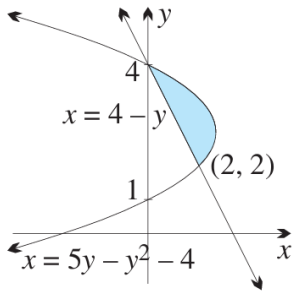
b



$A = \int_1^2 (3y - 2 - y^2)\,dy$

$= \frac{1}{6}$ u²

c

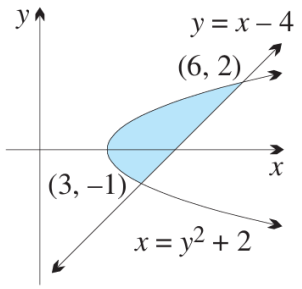


$A = \int_2^4 (5y - y^2 - 4 - 4 + y)\,dy$

$= \int_2^4 (6y - y^2 - 8)\,dy$

$= \frac{4}{3}$ u²

d



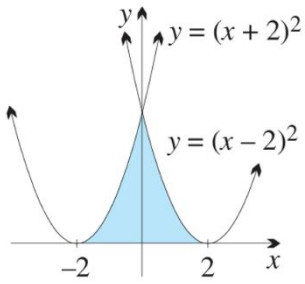
$A = \int_{-1}^2 (y + 4 - y^2 - 2)\,dy$

$= \int_{-1}^2 (y + 2 - y^2)\,dy$

$= \frac{9}{2}$ u²

3 Find the areas of the shaded regions. You will need to find two areas and add.

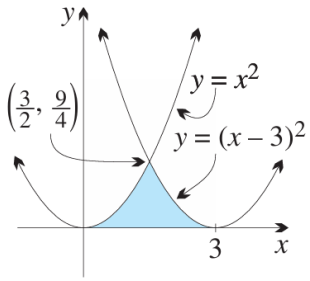
a



$\int_{-2}^0 (x-2)^2\,dx + \int_0^2 (x+2)^2\,dx$

$5\frac{1}{3}$ u²

b

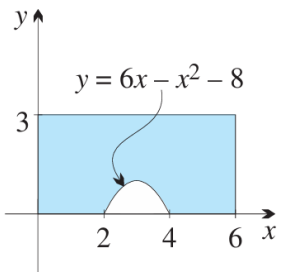


$\int_0^{\frac{3}{2}} x^2\,dx + \int_{\frac{3}{2}}^3 (x-3)^2\,dx$

$\frac{9}{4}$ u²

4 Find the areas of the shaded regions. You will need to find two areas and subtract.

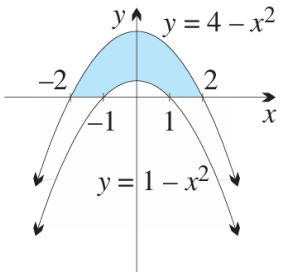
a



$3(6) - \int_2^4 6x - x^2 - 8\,dx$

$= 16\frac{2}{3}\text{ u}^2$

b



$\int_{-2}^{2} 4 - x^2\,dx - \int_{-1}^{1} 1 - x^2\,dx$

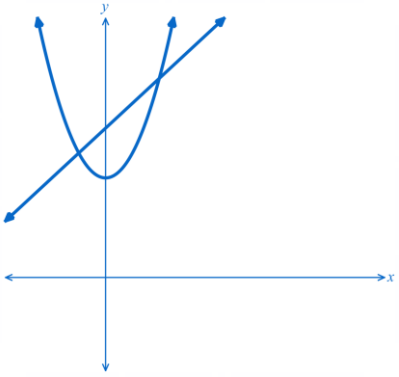
$= 9\frac{1}{3}\text{ u}^2$

5

a Show that $y = x^2 + 4$ and $y = x + 6$ intersect at $(-1, 5)$ and $(2, 8)$.

b Sketch the parabola and line, shading the enclosed region.

c Show that $\int_{-1}^{2}\big((x + 6) - (x^2 + 4)\big)\,dx = \int_{-1}^{2}(x - x^2 + 2)\,dx$, and evaluate.



$x^2 + 4 = x + 6$

$x^2 - x - 2 = 0$

$(x - 2)(x + 1) = 0$

$x = 2 \text{ or } x = -1$

$x = 2: y = 2 + 6 = 8 \rightarrow (2, 8)$

$x = -1: y = -1 + 6 = 5 \rightarrow (-1, 5)$

$\int_{-1}^{2}\big((x + 6) - (x^2 + 4)\big)\,dx = \int_{-1}^{2}(x - x^2 + 2)\,dx$

$= \left[\frac{x^2}{2} - \frac{x^3}{3} + 2x\right]_{-1}^{2}$

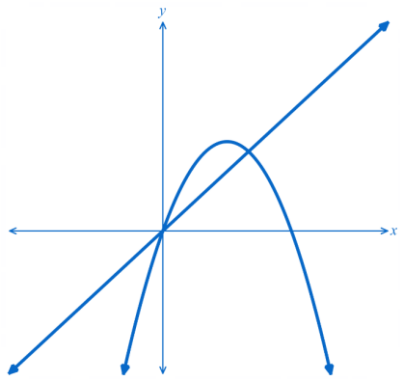
$= \frac{9}{2}\text{ u}^2$

6

a Show that $y = 3x - x^2 = x(3 - x)$ and $y = x$ intersect at $(0, 0)$ and $(2, 2)$.

b Sketch the parabola and line, shading the enclosed region.

c Show that $\int_0^2(3x - x^2 - x)\,dx = \int_0^2(2x - x^2)\,dx$, and evaluate.



$3x - x^2 = x$

$2x - x^2 = 0$

$x(2 - x) = 0$

$x = 0 \text{ or } x = 2$

$x = 0: y = 0 \rightarrow (0, 0)$

$x = 2: y = 2 \rightarrow (2, 2)$

$\int_0^2(3x - x^2 - x)\,dx = \int_0^2(2x - x^2)\,dx$

$= \left[x^2 - \frac{x^3}{3}\right]_0^2$

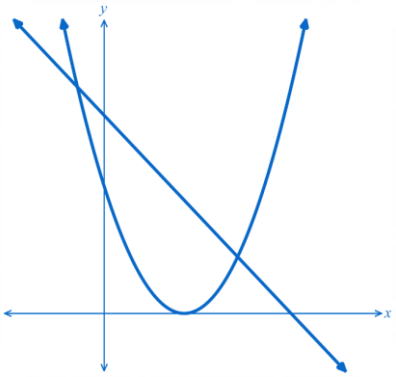
$= \frac{4}{3}\text{ u}^2$

7

a Show that $y = (x - 3)^2$ and $y = 14 - 2x$ intersect at $(-1, 16)$ and $(5, 4)$.

b Sketch the parabola and line, shading the enclosed region.

c Show that $\int_{-1}^{5}\left((14 - 2x) - (x - 3)^2\right) dx = \int_{-1}^{5}(4x + 5 - x^2)\, dx$, and evaluate.



$(x - 3)^2 = 14 - 2x$

$x^2 - 6x + 9 = 14 - 2x$

$x^2 - 4x - 5 = 0$

$(x - 5)(x + 1) = 0$

$x = 5 \text{ or } x = -1$

$x = 5: y = 14 - 10 = 4 \rightarrow (5, 4)$

$x = -1: y = 14 + 2 = 16 \rightarrow (-1, 16)$

$(14 - 2x) - (x - 3)^2 = 4x + 5 - x^2$

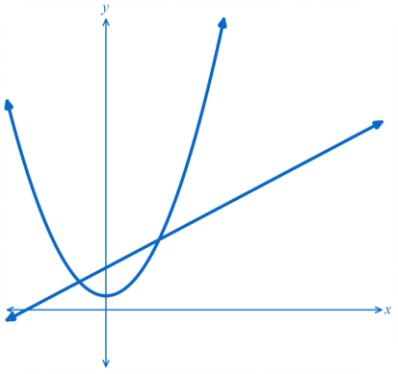
$\int_{-1}^{5}(4x + 5 - x^2)\, dx = \left[2x^2 + 5x - \frac{x^3}{3}\right]_{-1}^{5}$

$= \left(50 + 25 - \frac{125}{3}\right) - \left(2 - 5 + \frac{1}{3}\right)$

$= 36 \text{ u}^2$

8 Solve simultaneously, sketch, shade the enclosed region, and find its area for:

a $y = x + 3$ and $y = x^2 + 1$



$x + 3 = x^2 + 1$

$x^2 - x - 2 = 0$

$(x - 2)(x + 1) = 0$

$x = 2 \text{ or } x = -1$

Intersections: $(2, 5)$ and $(-1, 2)$

Line is above parabola on $[-1, 2]$.

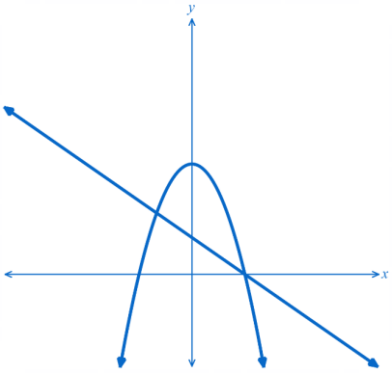
$\int_{-1}^{2}((x + 3) - (x^2 + 1))\, dx = \int_{-1}^{2}(x - x^2 + 2)\, dx$

$= \left[\frac{x^2}{2} - \frac{x^3}{3} + 2x\right]_{-1}^{2}$

$= \left(2 - \frac{8}{3} + 4\right) - \left(\frac{1}{2} + \frac{1}{3} - 2\right)$

$= \frac{9}{2} \text{ u}^2$

b $y = 9 - x^2$ and $y = 3 - x$



$9 - x^2 = 3 - x$

$x^2 - x - 6 = 0$

$(x - 3)(x + 2) = 0$

$x = 3 \text{ or } x = -2$

Intersections: $(3, 0)$ and $(-2, 5)$

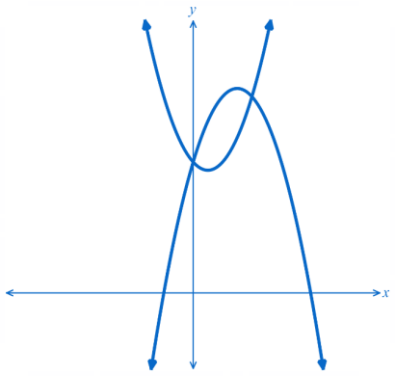
Parabola is above line on $[-2, 3]$.

$\int_{-2}^{3}((9 - x^2) - (3 - x))\, dx = \int_{-2}^{3}(6 + x - x^2)\, dx$

$= \left[6x + \frac{x^2}{2} - \frac{x^3}{3}\right]_{-2}^{3}$

$= \left(18 + \frac{9}{2} - 9\right) - \left(-12 + 2 + \frac{8}{3}\right) = \frac{125}{6} \text{ u}^2$

c $y = x^2 - x + 4$ and $y = -x^2 + 3x + 4$



$x^2 - x + 4 = -x^2 + 3x + 4$

$2x^2 - 4x = 0$

$2x(x - 2) = 0$

$x = 0$ or $x = 2$

Intersections: $(0, 4)$ and $(2, 6)$

Second parabola is above first on $[0, 2]$.

$$\int_0^2 ((-x^2 + 3x + 4) - (x^2 - x + 4))\,dx = \int_0^2 (4x - 2x^2)\,dx$$

$$= \left[2x^2 - \frac{2x^3}{3}\right]_0^2$$

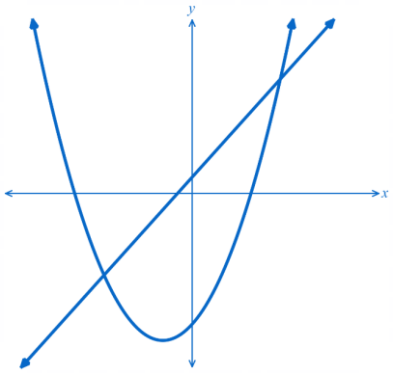
$$= 8 - \frac{16}{3} = \frac{8}{3}\text{ u}^2$$

9

a Show that $y = x^2 + 2x - 8$ and $y = 2x + 1$ intersect at $(3, 7)$ and $(-3, -5)$.

b Sketch both graphs, shading the enclosed region.

c Show that $\int_{-3}^{3} ((2x + 1) - (x^2 + 2x - 8))\,dx = \int_{-3}^{3} (9 - x^2)\,dx$.
Evaluate and find the area.



$x^2 + 2x - 8 = 2x + 1$

$x^2 - 9 = 0$

$(x - 3)(x + 3) = 0$

$x = 3$ or $x = -3$

$x = 3: y = 7 \rightarrow (3, 7)$

$x = -3: y = -5 \rightarrow (-3, -5)$

$(2x + 1) - (x^2 + 2x - 8) = 9 - x^2$

$$\int_{-3}^{3} (9 - x^2)\,dx = \left[9x - \frac{x^3}{3}\right]_{-3}^{3}$$

$$= (27 - 9) - (-27 + 9)$$

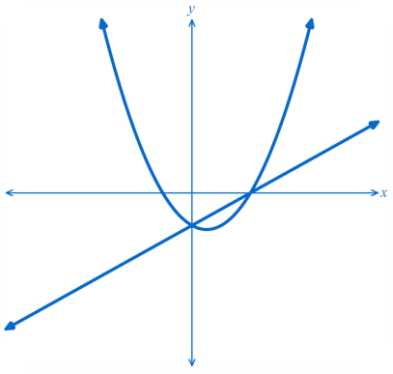
$$= 36\text{ u}^2$$

10

a Show that $y = x^2 - x - 2$ and $y = x - 2$ intersect at $(0, -2)$ and $(2, 0)$.

b Sketch both graphs, shading the enclosed region.

c Show that $\int_0^2 ((x - 2) - (x^2 - x - 2))\,dx = \int_0^2 (2x - x^2)\,dx$. Evaluate and find the area.



$x^2 - x - 2 = x - 2$

$x^2 - 2x = 0$

$x(x - 2) = 0$

$x = 0$ or $x = 2$

$x = 0: y = -2 \rightarrow (0, -2)$

$x = 2: y = 0 \rightarrow (2, 0)$

$(x - 2) - (x^2 - x - 2) = 2x - x^2$

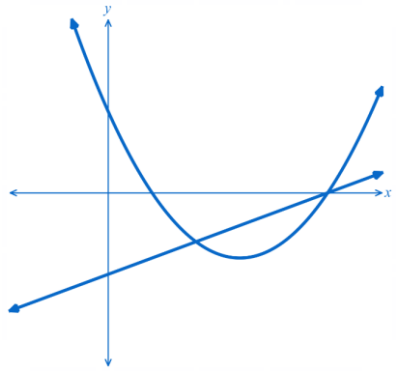
$$\int_0^2 (2x - x^2)\,dx = \left[x^2 - \frac{x^3}{3}\right]_0^2$$

$$= 4 - \frac{8}{3}$$

$$= \frac{4}{3}\text{u}^2$$

11 Solve simultaneously, sketch, shade the enclosed region, and find its area for:

a $y = x^2 - 6x + 5$ and $y = x - 5$



$x^2 - 6x + 5 = x - 5$

$x^2 - 7x + 10 = 0$

$(x - 2)(x - 5) = 0$

$x = 2$ or $x = 5$

Intersections: $(2, -3)$ and $(5, 0)$

Line is above parabola on $[2, 5]$.

$$\int_2^5 ((x - 5) - (x^2 - 6x + 5))\,dx = \int_2^5 (7x - 10 - x^2)\,dx$$

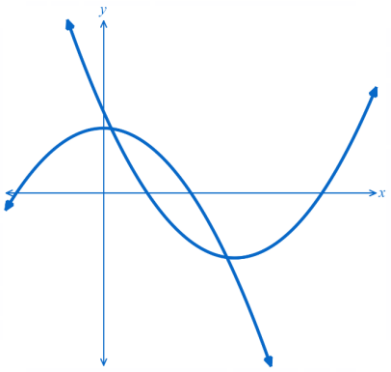
$$= \left[\frac{7x^2}{2} - 10x - \frac{x^3}{3}\right]_2^5$$

$$= \left(\frac{175}{2} - 50 - \frac{125}{3}\right) - \left(14 - 20 - \frac{8}{3}\right)$$

$$= \frac{25}{6} - \left(-\frac{26}{3}\right)$$

$$= \frac{9}{2}\text{ u}^2$$

b $y = -3x$ and $y = 4 - x^2$



$-3x = 4 - x^2$

$x^2 - 3x - 4 = 0$

$(x - 4)(x + 1) = 0$

$x = 4$ or $x = -1$

Intersections: $(4, -12)$ and $(-1, 3)$

Parabola is above line on $[-1, 4]$.

$$\int_{-1}^4 ((4 - x^2) - (-3x))\,dx = \int_{-1}^4 (4 + 3x - x^2)\,dx$$

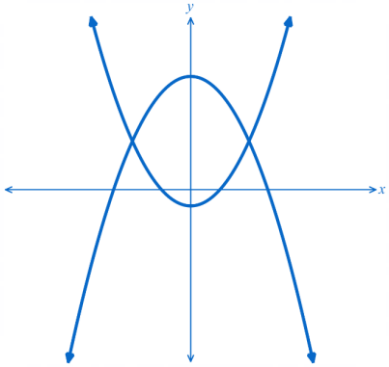
$$= \left[4x + \frac{3x^2}{2} - \frac{x^3}{3}\right]_{-1}^4$$

$$= \left(16 + 24 - \frac{64}{3}\right) - \left(-4 + \frac{3}{2} + \frac{1}{3}\right)$$

$$= \frac{56}{3} - \left(-\frac{13}{6}\right)$$

$$= \frac{125}{6}\text{ u}^2$$

c $y = x^2 - 1$ and $y = 7 - x^2$



$x^2 - 1 = 7 - x^2$

$2x^2 = 8$

$x^2 = 4$

$x = 2$ or $x = -2$

Intersections: $(2, 3)$ and $(-2, 3)$

Second curve is above first on $[-2, 2]$.

$$\int_{-2}^2 ((7 - x^2) - (x^2 - 1))\,dx = \int_{-2}^2 (8 - 2x^2)\,dx$$

$$= \left[8x - \frac{2x^3}{3}\right]_{-2}^2$$

$$= \left(16 - \frac{16}{3}\right) - \left(-16 + \frac{16}{3}\right)$$

$$= \frac{32}{3} + \frac{32}{3}$$

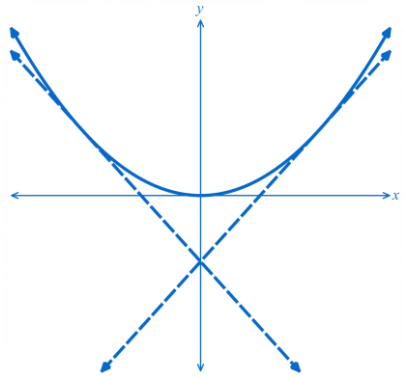
$$= \frac{64}{3}\text{ u}^2$$

12 Tangents are drawn to $x^2 = 8y$ at $A(4, 2)$ and $B(-4, 2)$.

a Draw a diagram noting symmetry about the y-axis.

b Find the equation of the tangent at A.

c Find the area bounded by the curve and the tangents.



$x^2 = 8y \rightarrow y = \frac{x^2}{8}$

$y' = \frac{x}{4}$

At $A(4, 2)$: $m = \frac{4}{4} = 1$

$y - 2 = 1(x - 4)$

$y = x - 2$

By symmetry, tangent at B: $y = -x - 2$

Tangents meet where $x - 2 = -x - 2$, so $x = 0, y = -2$.

Tangents meet at $(0, -2)$.

By symmetry, area $= 2 \times$ area from $x = 0$ to $x = 4$ between curve and tangent at A.

$$Area = 2\int_0^4 \left(\frac{x^2}{8} - (x - 2)\right) dx = 2\int_0^4 \left(\frac{x^2}{8} - x + 2\right) dx$$

$$= 2\left[\frac{x^3}{24} - \frac{x^2}{2} + 2x\right]_0^4$$

$$= 2\left(\frac{64}{24} - 8 + 8\right)$$

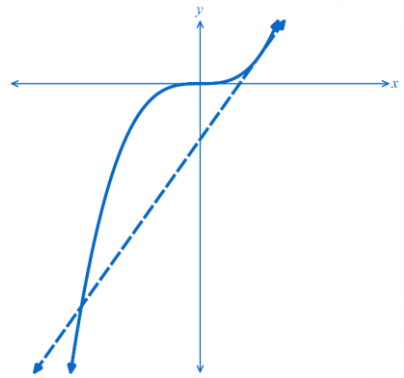
$$= \frac{16}{3} \text{ u}^2$$

13

a Show that the tangent to $y = x^3$ at $x = 2$ has equation $y = 12x - 16$.

b Show that the tangent and curve intersect again at $(-4, -64)$.

c Find the area enclosed between the curve and the tangent.



$y = x^3$

$y' = 3x^2$

$At\ x = 2: y = 8, y' = 12$

$y - 8 = 12(x - 2)$

$y = 12x - 16$

$x^3 = 12x - 16$

$x^3 - 12x + 16 = 0$

$(x - 2)^2$ is a factor (tangent touches at $x = 2$).

$x^3 - 12x + 16 = (x - 2)^2(x + 4)$

$x = -4: y = (-4)^3 = -64$

Intersects again at $(-4, -64)$.

$$\int_{-4}^{2} (x^3 - (12x - 16))\, dx = \int_{-4}^{2} (x^3 - 12x + 16)\, dx$$

$$= \left[\frac{x^4}{4} - 6x^2 + 16x\right]_{-4}^{2}$$

$$= (4 - 24 + 32) - (64 - 96 - 64)$$

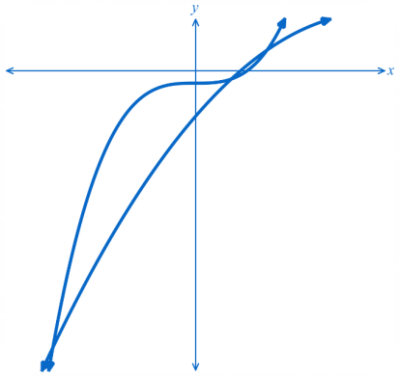
$$= 108 \text{ u}^2$$

14 Consider $y = x^3 - 3$ and $y = -x^2 + 10x - 11$.

a Show by substitution that the curves intersect where $x = -4, 1$ and 2.

b Sketch the curves showing intersection points.

c Find the area enclosed by the two curves.



$Set\ x^3 - 3 = -x^2 + 10x - 11$

$x^3 + x^2 - 10x + 8 = 0$

$x = -4: -64 + 16 + 40 + 8 = 0$

$x = 1: 1 + 1 - 10 + 8 = 0$

$x = 2: 8 + 4 - 20 + 8 = 0$

Intersection points:

$x = -4: y = (-4)^3 - 3 = -67 \rightarrow (-4, -67)$

$x = 1: y = 1 - 3 = -2 \rightarrow (1, -2)$

$x = 2: y = 8 - 3 = 5 \rightarrow (2, 5)$

$$\text{Area} = \int_{-4}^{1} \left((x^3 - 3) - (-x^2 + 10x - 11)\right) dx + \int_{1}^{2} \left((-x^2 + 10x - 11) - (x^3 - 3)\right) dx$$

$$= \int_{-4}^{1} (x^3 + x^2 - 10x + 8)\, dx + \int_{1}^{2} (-x^3 - x^2 + 10x - 8)\, dx$$

$$\int_{-4}^{1} (x^3 + x^2 - 10x + 8)\, dx = \left[\frac{x^4}{4} + \frac{x^3}{3} - 5x^2 + 8x\right]_{-4}^{1}$$

$$= \frac{43}{12} - \left(-\frac{208}{3}\right) = \frac{875}{12}$$

$$\int_{1}^{2} (-x^3 - x^2 + 10x - 8)\, dx = -\left[\frac{x^4}{4} + \frac{x^3}{3} - 5x^2 + 8x\right]_{1}^{2}$$

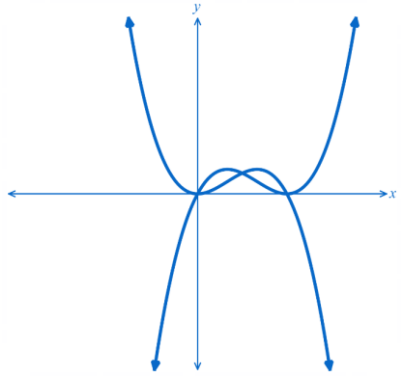
$$= -\left(\frac{32}{12} - \frac{43}{12}\right) = \frac{11}{12}$$

$$\text{Area} = \frac{875}{12} + \frac{11}{12} = \frac{443}{6} u^2$$

15

a Find the points of intersection of $y = x^2(1 - x)$ and $y = x(1 - x)^2$.

b Find the area bounded by the two curves.



$$x^2(1 - x) = x(1 - x)^2$$
$$x^2(1 - x) - x(1 - x)^2 = 0$$
$$x(1 - x)[x - (1 - x)] = 0$$
$$x(1 - x)(2x - 1) = 0$$
$$x = 0, x = 1, x = \frac{1}{2}$$
$$x = 0: y = 0 \rightarrow (0, 0)$$
$$x = \frac{1}{2}: y = \left(\frac{1}{4}\right)\left(\frac{1}{2}\right) = \frac{1}{8} \rightarrow \left(\frac{1}{2}, \frac{1}{8}\right)$$
$$x = 1: y = 0 \rightarrow (1, 0)$$

Let $f(x) = x^2(1 - x)$ and $g(x) = x(1 - x)^2$

$f(x) - g(x) = x(1 - x)(x - (1 - x)) = x(1 - x)(2x - 1)$

On $\left[0, \frac{1}{2}\right]$: $(2x - 1) < 0$, so $f < g$ (second curve above)

On $\left[\frac{1}{2}, 1\right]$: $(2x - 1) > 0$, so $f > g$ (first curve above)

By symmetry, both regions have equal area.

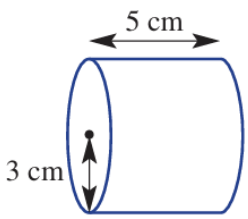
$$\text{Area} = 2\int_0^{\frac{1}{2}} (x(1 - x)^2 - x^2(1 - x))\, dx$$
$$= 2\int_0^{\frac{1}{2}} x(1 - x)(1 - 2x)\, dx$$
$$= 2\int_0^{\frac{1}{2}} (x - 3x^2 + 2x^3)\, dx$$
$$= 2\left[\frac{x^2}{2} - x^3 + \frac{x^4}{2}\right]_0^{\frac{1}{2}}$$
$$= 2\left(\frac{1}{8} - \frac{1}{8} + \frac{1}{32}\right)$$
$$= \frac{1}{16}\ \text{u}^2$$

Volumes of Rotation

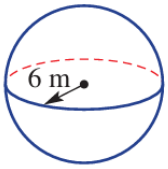
Review

☐ Calculate volume of solids

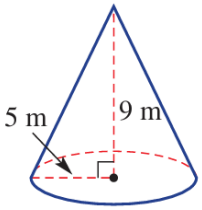
a



b



c



☐ Evaluate definite Integrals

a

$$\int_1^4 x^2 - 3\,dx$$

12

b

$$\int_1^4 e^{2x}\,dx$$

$\frac{e^8 - e^2}{2}$

c

$$\int_1^4 (x-3)^2\,dx$$

3

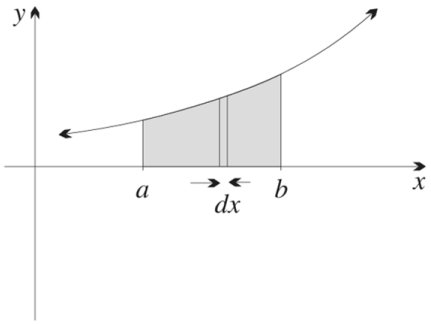
d

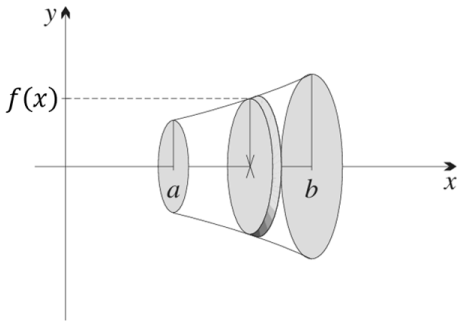
$$\int_1^4 (x^2-3)^2\,dx$$

$\frac{528}{5}$

Investigation Volume of solids of revolution

When you rotate the area under a curve around an axis, you create a **solid of revolution**.



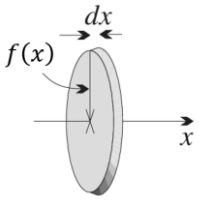


This solid is made up of thin circular slices.

The radius of each slice is: ……

The thickness of each slice is: ……

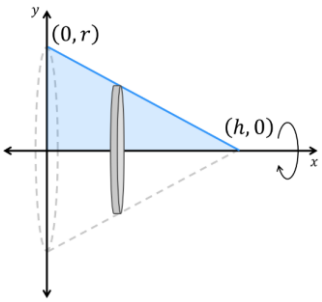
So, the volume of each slice is: ……

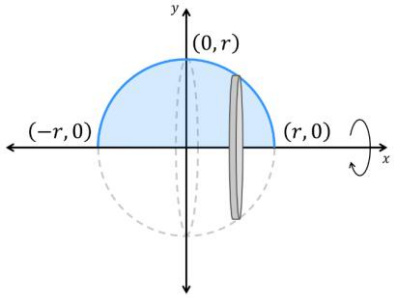


We add up every slice between $x = a$ and $x = b$, which is equivalent to integrating:

$$V = \int_a^b \pi[f(x)]^2\,dx$$
$$= \pi \int_a^b [f(x)]^2\,dx$$

Let's use this formula to derive the volumes of a cone and sphere:

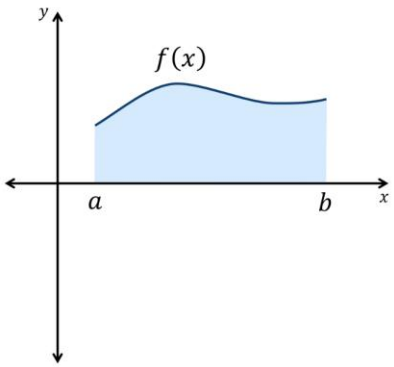


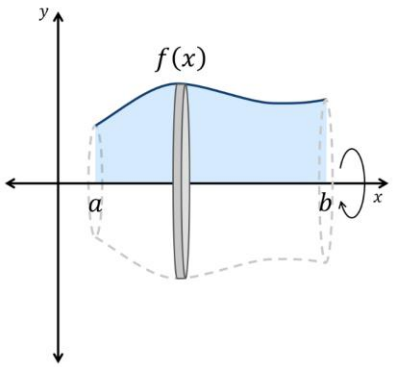


ⓘ Volume of Solids Rotated about x axis

Rotating $y = f(x)$ about x-axis between $x = a$ and $x = b$:

$$V = \pi \int_a^b y^2 \, dx$$





Example Calculate volumes of rotation about x-axis

Find the volume of the solid formed when the line $y = 3x - 2$ is rotated about the x-axis from $x = 1$ to $x = 2$

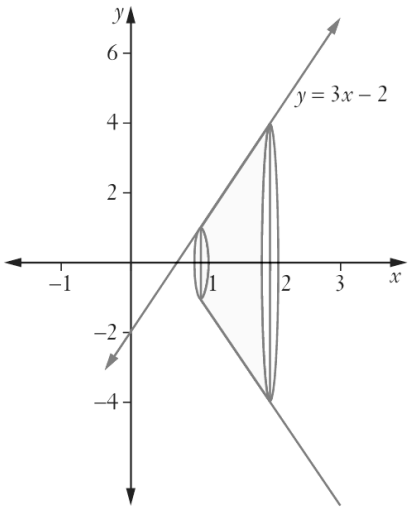
$$V = \pi \int_1^2 (3x - 2)^2 \, dx$$

$$= \pi \left[\frac{(3x-2)^3}{9} \right]_1^2$$

$$= \frac{\pi}{9} \left[(3x-2)^3 \right]_1^2$$

$$= \frac{\pi}{9} (4^3 - 1^3)$$

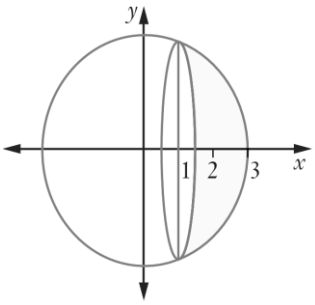
$$= 7\pi \text{ units}^3$$



Guided Practice

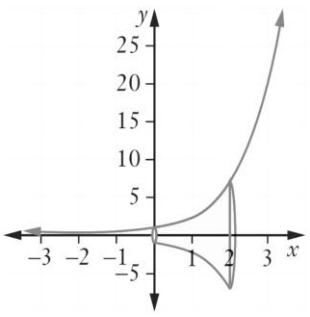
Find the volume of the solid formed by rotating:

a $x^2 + y^2 = 9$ between $x = 1$ and $x = 3$.



$\frac{28\pi}{3}$ u³

b $y = e^x$ between $x = 0$ and $x = 2$.

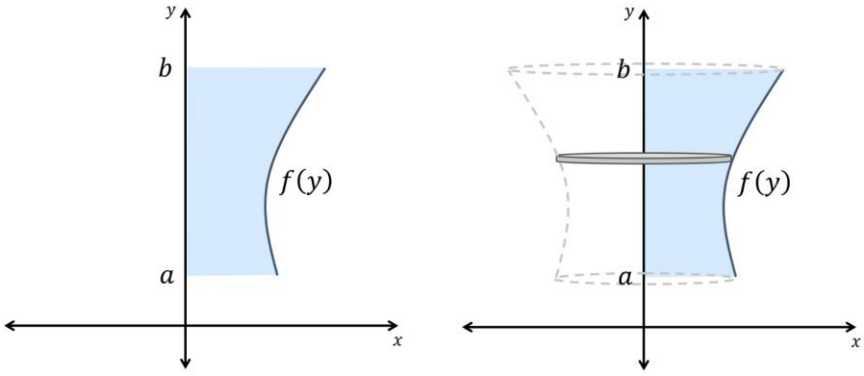


$\frac{\pi}{2}(e^4 - 1)$ u³

ⓘ Volume of Solids Rotated about y axis

Rotating $x = f(y)$ about y-axis between $x = a$ and $x = b$:

$$V = \pi \int_a^b x^2 \, dy$$



Example Calculate volumes of rotation about y-axis

Find the volume of the solid formed when the line $y = x^2 - 1$ is rotated about the y-axis from $y = -1$ to $y = 3$

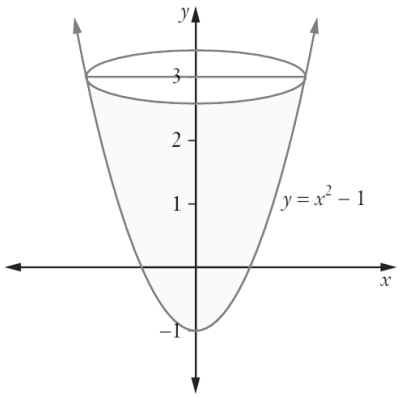
$x^2 = y + 1$

$$V = \pi \int_{-1}^{3} y + 1 \, dy$$

$$= \pi \left[\frac{y^2}{2} + y \right]_{-1}^{3}$$

$$= \pi \left(\left(\frac{9}{2} + 3 \right) - \left(\frac{1}{2} - 1 \right) \right)$$

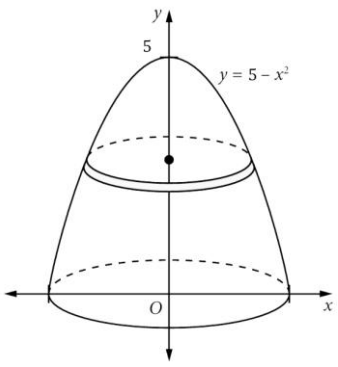
$$= 8\pi \text{ units}^3$$



Guided Practice

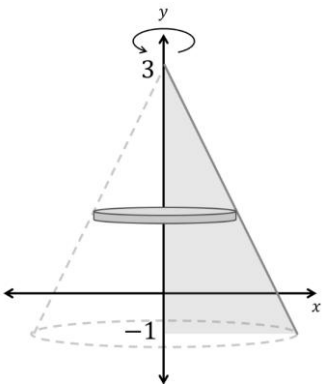
Find the volume of the solid formed by rotating these regions about the y axis:

a $y = 5 - x^2$ between $y = 0$ and $y = 5$.



$\frac{25\pi}{2}$ u³

b $y = 3 - 2x$ between $y = -1$ and $y = 3$.



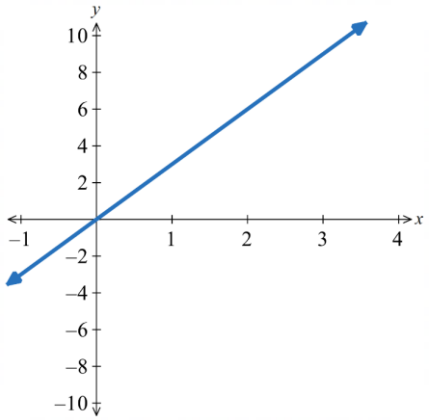
$\frac{16\pi}{3}$ u³

1

a Consider the region bounded by the line $y = 3x$ and the x-axis between $x = 0$ and $x = 3$.

b When this region is rotated about the x-axis, a right circular cone is formed.
Find the radius and height of the cone and hence find its volume using $V = \frac{1}{3}\pi r^2 h$

c Evaluate $\pi \int_0^3 9x^2\,dx$ to check your answer.



At $x = 3$: $y = 9$, so $r = 9, h = 3$.

$V = \frac{1}{3}\pi(9^2)(3) = 81\pi$ u³

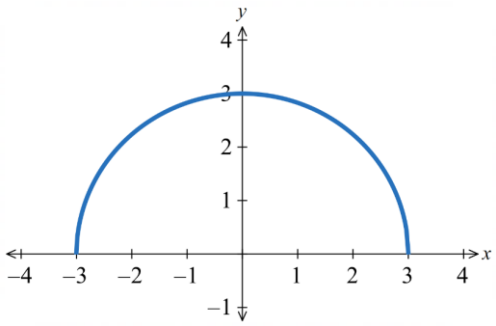
$\pi \int_0^3 9x^2\,dx = \pi[3x^3]_0^3 = 81\pi$ u³

2

a Consider the region bounded by the curve $y = \sqrt{9 - x^2}$ and the x-axis between $x = -3$ and $x = 3$.

b When this region is rotated about the x-axis, a sphere is formed.
Find the radius of the sphere and hence find its volume using $V = \frac{4}{3}\pi r^3$

c Evaluate $\pi \int_{-3}^3 (9 - x^2)\,dx$ to check your answer.



$r = 3$.

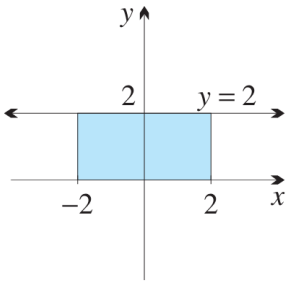
$V = \left(\frac{4}{3}\right)\pi(3^3) = 36\pi$ u³

$\pi \int_{-3}^3 (9 - x^2)\,dx = \pi\left[9x - \frac{x^3}{3}\right]_{-3}^3$

$= \pi(18 + 18) = 36\pi$ u³

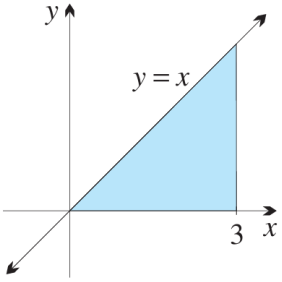
3 Calculate the volume generated when each shaded region is rotated about the x-axis.

a



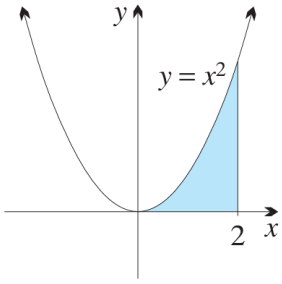
16π u³

b



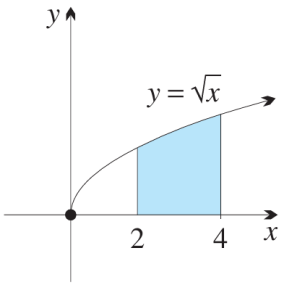
9π u³

c



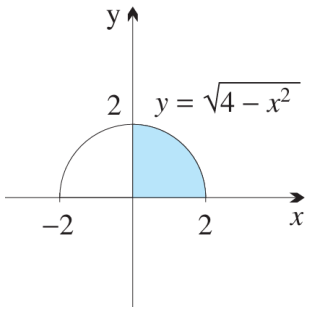
$\frac{32\pi}{5}$ u³

d



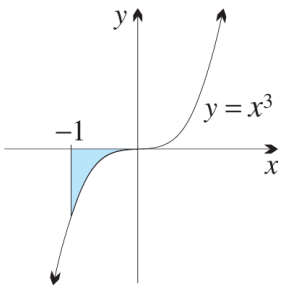
6π u³

e



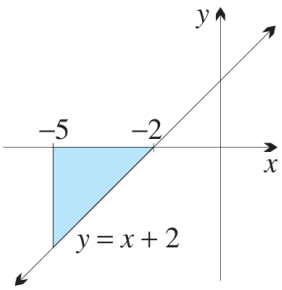
$\frac{16\pi}{3}$ u³

f



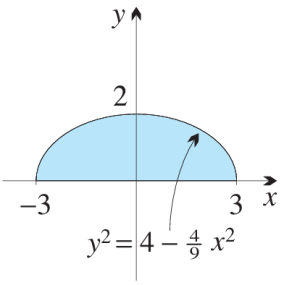
$\frac{\pi}{7}$ u³

g



9π u³

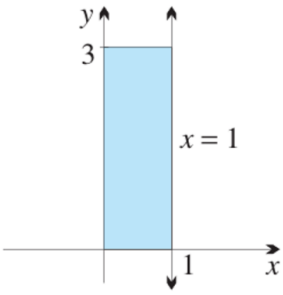
h



16π u³

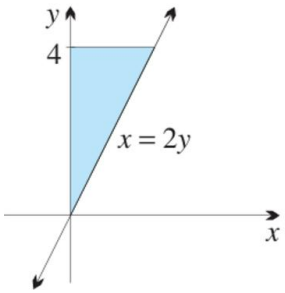
4 Calculate the volume generated when each shaded region is rotated about the y-axis.

a



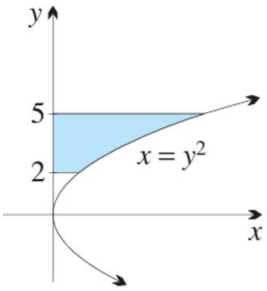
3π u³

b



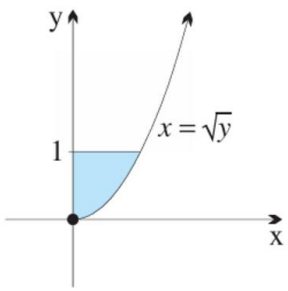
$\frac{256\pi}{3}$ u³

c



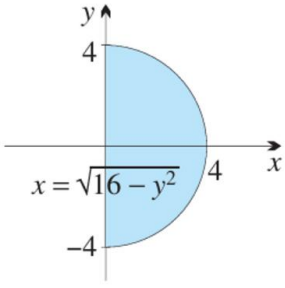
8π u³

d



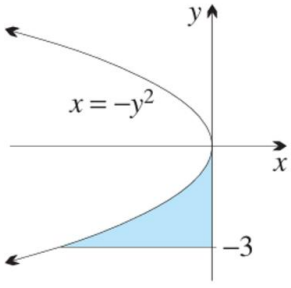
$\frac{\pi}{2}$ u³

e



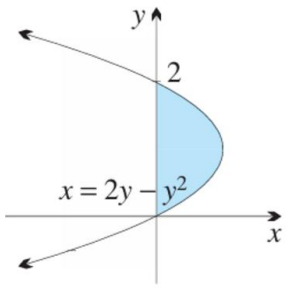
$\frac{256\pi}{3}$ u³

f



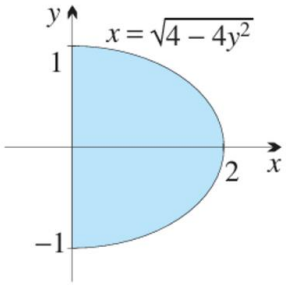
$\frac{243\pi}{5}$ u³

g



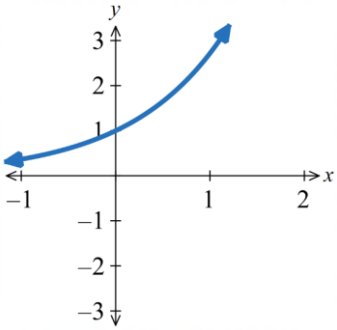
$\frac{16\pi}{5}$ u³

h



$\frac{16\pi}{3}$ u³

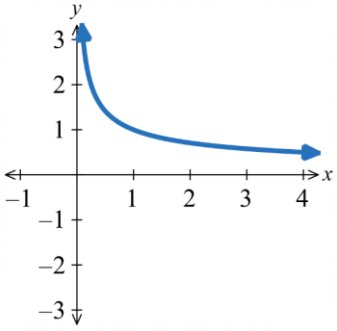
5 The region between the curve $y = e^x$ and the x-axis from $x = 0$ to $x = 1$ is rotated about the x-axis. Find the volume of the solid generated.



$V = \pi \int_0^1 e^{2x}\,dx = \pi\left[\frac{e^{2x}}{2}\right]_0^1$

$= \left(\frac{\pi}{2}\right)(e^2 - 1)\text{u}^3$

6 Find the volume generated when the region between the curve $y = \frac{1}{\sqrt{x}}$ and the x-axis, from $x = 2$ to $x = 4$, is rotated about the x-axis.

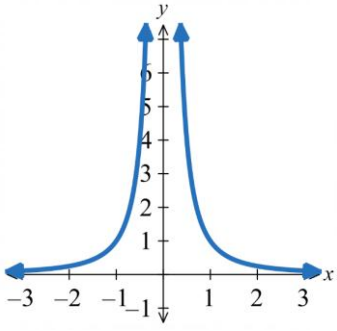


$y^2 = \frac{1}{x}$

$V = \pi \int_2^4 \left(\frac{1}{x}\right) dx = \pi[\ln x]_2^4$

$= \pi \ln 2\ \text{u}^3$

7 The region between the curve $y = \frac{1}{x^2}$ (a truncus) and the y-axis from $y = 1$ to $y = 6$ is rotated about the y-axis. Calculate the exact volume of the solid formed.



$x^2 = \frac{1}{y}$

$V = \pi \int_1^6 \left(\frac{1}{y}\right) dy = \pi[\ln y]_1^6$

$= \pi \ln 6\ \text{u}^3$

8

a Write $\tan^2 x$ in terms of $\sec^2 x$.

b The region bounded by the curve $y = \tan x$, the x-axis and the vertical line $x = \frac{\pi}{3}$ is rotated about the x-axis. Find the volume of the solid generated.

$$a\ \tan^2 x = \sec^2 x - 1$$

$$V = \pi \int_0^{\frac{\pi}{3}} (\sec^2 x - 1)\, dx = \pi[\tan x - x]_0^{\frac{\pi}{3}}$$

$$= \pi\left(\sqrt{3} - \frac{\pi}{3}\right) \text{u}^3$$

9

a Write $\sin^2 x$ in terms of $\cos 2x$.

b The region bounded by the curve $y = \sin x$, the x-axis and the vertical line $x = \frac{\pi}{2}$ is rotated about the x-axis. Find the volume of the solid generated.

$$\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos 2x$$

$$V = \pi \int_0^{\frac{\pi}{2}} \left(\frac{1}{2} - \left(\frac{1}{2}\right)\cos 2x\right) dx = \pi\left[\frac{x}{2} - \frac{\sin 2x}{4}\right]_0^{\frac{\pi}{2}}$$

$$= \frac{\pi^2}{4} \text{u}^3$$

10 Find the volume of the solid formed by rotating the region with the given boundaries about the x-axis. (A sketch of each region will be needed.)

a $y = x + 3, x = 3, x = 5, y = 0$

$$V = \pi \int_3^5 (x + 3)^2\, dx$$

$$= \pi\left[\frac{(x + 3)^3}{3}\right]_3^5$$

$$= \frac{296\pi}{3} \text{u}^3$$

b $y = 1 + \sqrt{x}, x = 1, x = 4, y = 0$

$$V = \pi \int_1^4 \left(1 + 2x^{\frac{1}{2}} + x\right) dx$$

$$= \pi\left[x + \left(\frac{4}{3}\right)x^{\frac{3}{2}} + \frac{x^2}{2}\right]_1^4$$

$$= \frac{119\pi}{6} \text{u}^3$$

c $y = 5x - x^2, y = 0$

Zeros at $x = 0, 5$.

$$V = \pi \int_0^5 (25x^2 - 10x^3 + x^4)\, dx$$

$$= \pi\left[\frac{25x^3}{3} - \left(\frac{5}{2}\right)x^4 + \frac{x^5}{5}\right]_0^5$$

$$= \frac{625\pi}{6} \text{u}^3$$

d $y = x^3 - x, y = 0$

Zeros at $x = -1, 0, 1$; $(x^3 - x)^2$ is even, so integrate on [0, 1] and double.

$$V = 2\pi \int_0^1 (x^6 - 2x^4 + x^2)\, dx$$

$$= 2\pi\left[\frac{x^7}{7} - \frac{2x^5}{5} + \frac{x^3}{3}\right]_0^1$$

$$= \frac{16\pi}{105} \text{u}^3$$

11 Find the volume of the solid formed by rotating the region with the given boundaries about the y-axis. (A sketch of each region will be needed.)

a $x = y - 2, y = 1, x = 0$

$x = 0$ when $y = 2$; radius $= 2 - y$ for $y \in [1, 2]$.

$$V = \pi \int_1^2 (2 - y)^2 \, dy$$
$$= \pi \left[-\frac{(2 - y)^3}{3} \right]_1^2$$
$$= \frac{\pi}{3} \text{u}^3$$

b $x = y^2 + 1, y = 0, y = 1, x = 0$

$$V = \pi \int_0^1 (y^2 + 1)^2 \, dy$$
$$= \pi \int_0^1 (y^4 + 2y^2 + 1) \, dy$$
$$= \pi \left[\frac{y^5}{5} + \frac{2y^3}{3} + y \right]_0^1$$
$$= \frac{28\pi}{15} u^3$$

c $x = y(y - 3), x = 0$

$x = y(y - 3) \leq 0$ on $(0, 3)$; radius $= y(3 - y)$.

$$V = \pi \int_0^3 y^2 (3 - y)^2 \, dy$$
$$= \pi \int_0^3 (9y^2 - 6y^3 + y^4) \, dy$$
$$= \pi \left[3y^3 - \left(\frac{3}{2}\right) y^4 + \frac{y^5}{5} \right]_0^3$$
$$= \frac{81\pi}{10} u^3$$

d $y = 1 - x^2, y = 0$

$x^2 = 1 - y, y$ from 0 to 1.

$$V = \pi \int_0^1 (1 - y) \, dy$$
$$= \pi \left[y - \frac{y^2}{2} \right]_0^1$$
$$= \frac{\pi}{2} \text{u}^3$$

12 A vat is designed by rotating about the x-axis the region between the curve $y = 1 + \frac{1}{x}$ and the x-axis from $x = \frac{1}{2}$ to $x = 3$. Show that its volume is $\left(\frac{\pi}{6}\right)(25 + 12 \ln 6) \text{ u}^3$.

$$V = \pi \int_{\frac{1}{2}}^3 \left(1 + \frac{2}{x} + \frac{1}{x^2}\right) dx = \pi \left[x + 2 \ln x - \frac{1}{x} \right]_{\frac{1}{2}}^3$$
$$= \pi \left(\left(\frac{8}{3} + 2 \ln 3 \right) - \left(-\frac{3}{2} - 2 \ln 2 \right) \right)$$
$$= \frac{\pi}{6} (25 + 12 \ln 6)$$

13 A metal stud is created by rotating about the x-axis the region between the curve $y = e^x - e^{-x}$ and the x-axis from $x = 0$ to $x = \frac{1}{2}$. Show that the volume of the stud is $\left(\frac{\pi}{2}\right)(e - 2 - e^{-1})$ u^3.

$$y^2 = e^{2x} - 2 + e^{-2x}$$

$$V = \pi \int_0^{\frac{1}{2}} (e^{2x} - 2 + e^{-2x})\, dx$$

$$= \pi \left[\frac{e^{2x}}{2} - 2x - \frac{e^{-2x}}{2}\right]_0^{\frac{1}{2}}$$

$$= \pi \left(\frac{e}{2} - 1 - \frac{e^{-1}}{2}\right) = \frac{\pi}{2}(e - 2 - e^{-1})$$

14 A champagne flute is designed by rotating about the x-axis the region between the curve $y = 4 + 4\sin\left(\frac{x}{4}\right)$ and the x-axis from $x = 4\pi$ to $x = 6\pi$. Find, correct to 4 significant figures, the capacity of the flute if 1 unit = 1 cm.

$$y^2 = 16\left(1 + \sin\left(\frac{x}{4}\right)\right)^2$$

$$= 16\left(1 + 2\sin\left(\frac{x}{4}\right) + \sin^2\left(\frac{x}{4}\right)\right)$$

Using double angle rule: $\sin^2\left(\frac{x}{4}\right) = \frac{1}{2} - \frac{1}{2}\cos\left(\frac{x}{2}\right)$

$$= 16\left(\frac{3}{2} + 2\sin\left(\frac{x}{4}\right) - \frac{1}{2}\cos\left(\frac{x}{2}\right)\right)$$

$$V = 16\pi \int_{4\pi}^{6\pi} \left(\frac{3}{2} + 2\sin\left(\frac{x}{4}\right) - \frac{1}{2}\cos\left(\frac{x}{2}\right)\right) dx$$

$$= 16\pi \left[\frac{3x}{2} - 8\cos\left(\frac{x}{4}\right) - \sin\left(\frac{x}{2}\right)\right]_{4\pi}^{6\pi}$$

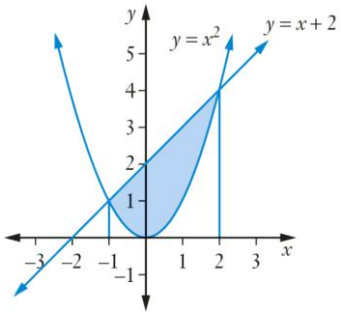
$$= 16\pi(3\pi - 8) \approx 71.62 \text{ mL}$$

Rotating Area between Two Curves

Review

☐ Calculate area between two curves

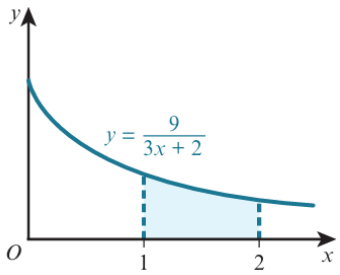
Find the area enclosed between the curve $y = x^2$ and the line $y = x + 2$



4.5 u²

☐ Calculate volume of solid of revolution

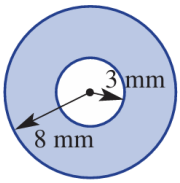
Rotate the given region about the x axis.



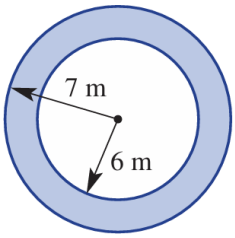
$\frac{81\pi}{40}$ u³

☐ Calculate area of an annulus

a

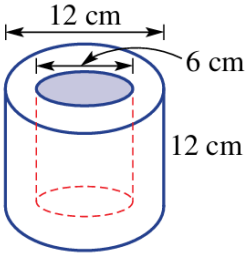


b

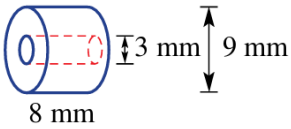


☐ Calculate volume of composite solid

a

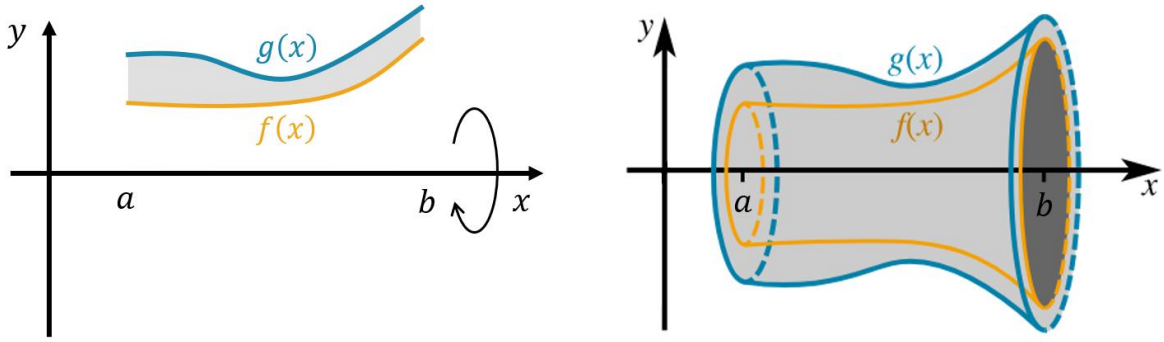


b



Investigation Volume of solids of revolution: area between curves

When you rotate the area between two curves around an axis, you create a hollow solid:



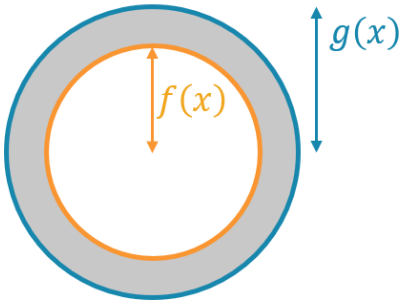
The cross-section of this solid is an **annulus**. Consider $g(x) \geq f(x)$:

The outer radius is: ……

The inner radius is: ……

The area of the annulus is: ……

The volume of a slice is: ……



We add up every slice between $x = a$ and $x = b$, which is equivalent to integrating:

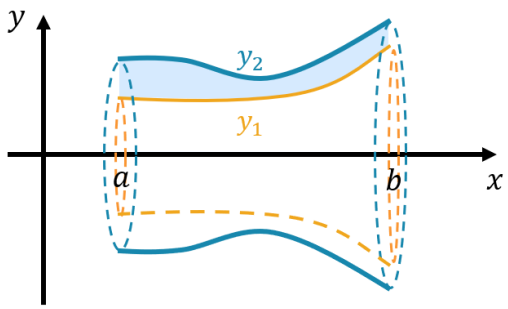
$$V = \int_a^b \pi([g(x)]^2 - [f(x)]^2)\,dx$$

$$V = \pi \int_a^b [g(x)]^2 - [f(x)]^2\,dx$$

ⓘ Volumes of Revolution: Area between two curves

Rotation about x axis

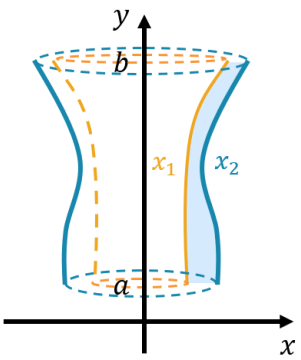
$y_2 \geq y_1$



$$V = \pi \int_a^b (y_2^2 - y_1^2)\, dx$$

Rotation about y axis

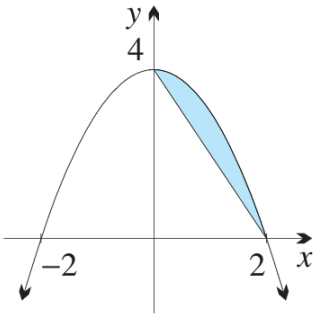
$x_2 \geq x_1$



$$V = \pi \int_a^b (x_2^2 - x_1^2)\, dy$$

Example Calculate volume of solids of revolution: area between curves

The area between the curve $y = 4 - x^2$ and the line $y = 4 - 2x$ between $x = 0$ and $x = 2$ are rotated about the x-axis. Find the volume generated.



$$V = \pi \int_0^2 (y_2^2 - y_1^2)\, dx \quad \text{where } y_2 = 4 - x^2 \text{ and } y_1 = 4 - 2x$$

$$= \pi \int_0^2 ((4 - x^2)^2 - (4 - 2x)^2)\, dx$$

$$= \pi \int_0^2 (x^4 - 12x^2 + 16x)\, dx$$

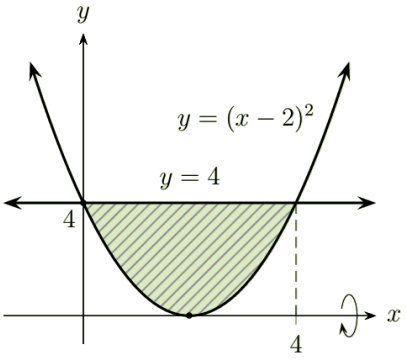
$$= \pi \left[\frac{x^5}{5} - 4x^3 + 9x^2\right]_0^2$$

$$= \pi \left(\frac{32}{5} - 32 + 32 - (0 - 0 + 0)\right)$$

$$= \frac{32\pi}{5} \text{ u}^3$$

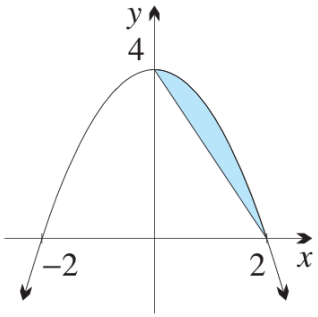
Guided Practice

a The shaded region bounded by $y = (x-2)^2$ and $y = 4$ between $x = 0$ and $x = 4$ is rotated about the x axis. Find the volume of this solid.



$\frac{256\pi}{5}$ u³

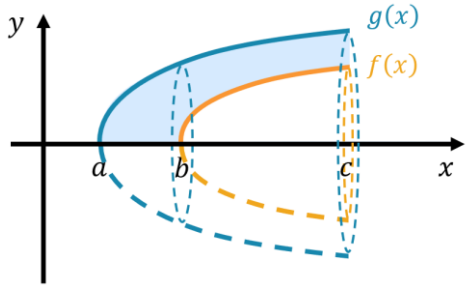
b The area between $y = 4 - x^2$ and $y = 4 - 2x$ between $y = 0$ and $y = 4$ are rotated about the y-axis. Find the volume generated.



$\frac{8\pi}{3}$ u³

Investigation Volume of solids of revolution: further area between curves

If the domain of the two functions are different, you must use two separate integrals.



Method 1: Split the region

Where both curves exist:

$$V_1 = \pi \int_b^c [g(x)]^2 - [f(x)]^2 \, dx$$

Where only one curve exists:

$$V_2 = \pi \int_a^b [g(x)]^2 \, dx$$

$$V_{\text{total}} = V_1 + V_2$$

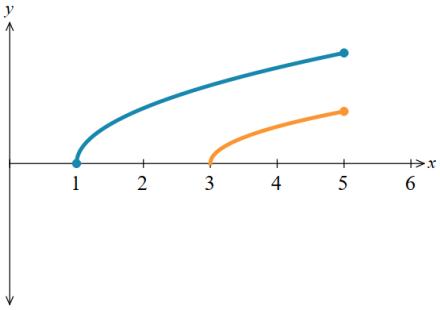
Method 2: Outer volume – Inner volume

$$V_{\text{outer}} = \pi \int_{\boldsymbol{a}}^{\boldsymbol{c}} [g(x)]^2 \, dx$$

$$V_{\text{inner}} = \pi \int_{\boldsymbol{b}}^{\boldsymbol{c}} [f(x)]^2 \, dx$$

$$V_{\text{total}} = V_{\text{outer}} - V_{\text{inner}}$$

The area bounded by $y = \sqrt{x-1}$ and $y = \sqrt{x-3}$ between $x = 1$ and $x = 5$ are rotated about the x axis. Find the volume of the solid.



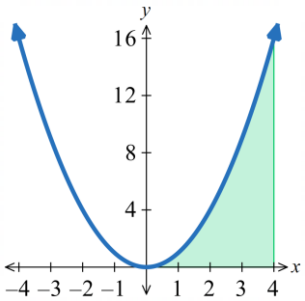
Method 1

6π u^3

Method 2

6π u^3

1 The region R is bounded by the parabola $y = x^2$ and the x-axis from $x = 0$ to $x = 4$.
We will calculate the volume of this region rotated around the y-axis by:



a Find the **outer volume:** cylinder formed when the region between the line $x = 4$ and the y-axis from $y = 0$ to $y = 16$ is rotated about the y-axis.

$V = \pi(4^2)(16) = 256\pi$ u³

b Find the **inner volume**: the region between the parabola $y = x^2$ and the y-axis from $y = 0$ to $y = 16$ is rotated about the y-axis.

x² = y; rotating about y-axis:

$V = \pi \int_0^{16} y\,dy = \pi \left[\frac{y^2}{2}\right]_0^{16} = 128\pi$ u³

c Hence use parts **a** and **b** to find the volume of the solid formed when R is rotated about the y-axis.

$V = 256\pi - 128\pi = 128\pi$ u³

d Calculate the same volume using $V = \pi \int_a^b (x_2^2 - x_1^2)\,dy$

$V = \pi \int_0^{16} 4^2 - y\,dy$

$= \pi \left[16y - \frac{y^2}{2}\right]_0^{16}$

$= 128\pi$ u³

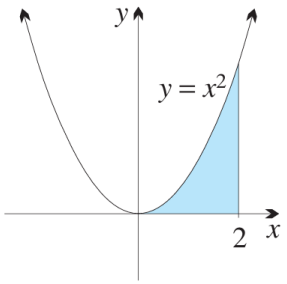
2 Find the volume of the solid generated by rotating each region about:

i the x-axis

ii the y-axis

In some cases, a subtraction of volumes will be necessary.

a



i

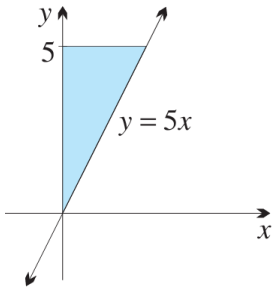
$$V = \pi \int_0^2 x^4\,dx$$
$$= \pi \left[\frac{x^5}{5}\right]_0^2$$
$$= \frac{32\pi}{5}\text{u}^3$$

ii

Outer radius 2, inner radius $\sqrt{y}$, y from 0 to 4:

$$V = \pi \int_0^4 (4 - y)\,dy$$
$$= \pi \left[4y - \frac{y^2}{2}\right]_0^4$$
$$= 8\pi\ u^3$$

b



i

Outer radius 5, inner radius $5x$, x from 0 to 1:

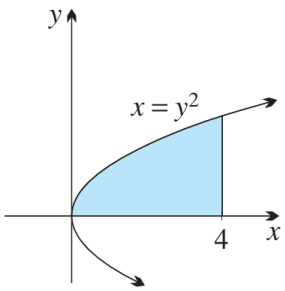
$$V = \pi \int_0^1 (25 - 25x^2)\,dx$$
$$= 25\pi \left[x - \frac{x^3}{3}\right]_0^1$$
$$= \frac{50\pi}{3}\text{u}^3$$

ii

$x = \frac{y}{5}$, y from 0 to 5:

$$V = \pi \int_0^5 \left(\frac{y}{5}\right)^2 dy$$
$$= \left(\frac{\pi}{25}\right)\left[\frac{y^3}{3}\right]_0^5 = \frac{5\pi}{3}\text{u}^3$$

c



i

$y = \sqrt{x}$, so $y^2 = x$:

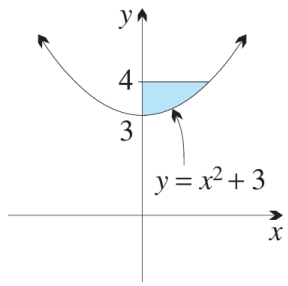
$$V = \pi \int_0^4 x\,dx$$
$$= \pi \left[\frac{x^2}{2}\right]_0^4$$
$$= 8\pi\ \text{u}^3$$

ii

Outer radius 4, inner radius y^2, y from 0 to 2:

$$V = \pi \int_0^2 (16 - y^4)\,dy$$
$$= \pi \left[16y - \frac{y^5}{5}\right]_0^2$$
$$= \frac{128\pi}{5}\text{u}^3$$

d



i

Outer radius 4, inner radius $x^2 + 3$, x from 0 to 1:

$$V = \pi \int_0^1 (16 - (x^2 + 3)^2)\,dx$$
$$= \pi \int_0^1 (7 - 6x^2 - x^4)\,dx$$
$$= \pi \left[7x - 2x^3 - \frac{x^5}{5}\right]_0^1 = \frac{24\pi}{5}\text{u}^3$$

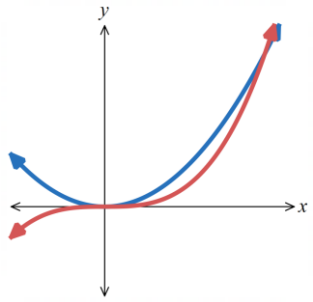
ii

$x^2 = y - 3$, y from 3 to 4:

$$V = \pi \int_3^4 (y - 3)\,dy$$
$$= \pi \left[\frac{(y-3)^2}{2}\right]_3^4 = \frac{\pi}{2}\text{u}^3$$

3

a Consider the region bounded by the curves $y = x^2$ and $y = x^3$.
Find the points of intersection.



$x^2 = x^3$

$x^2(x-1) = 0$

Intersections at $x = 0, x = 1$;

$\therefore (0,0)$ and $(1,1)$

b Find the volume of the solid generated when this region is rotated about:

iii the x-axis

$x^2 > x^3$ on $x \in (0,1)$.
Outer radius x^2, inner radius x^3:

$$V = \pi \int_0^1 (x^4 - x^6)\,dx = \pi \left[\frac{x^5}{5} - \frac{x^7}{7}\right]_0^1$$
$$= \frac{2\pi}{35}\text{u}^3$$

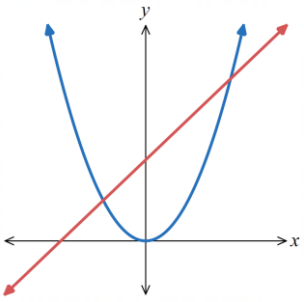
iv the y-axis

$y = x^2 \to x = y^{\frac{1}{2}}$;
$y = x^3 \to x = y^{\frac{1}{3}}$.
For $y \in (0,1), y^{\frac{1}{3}} > y^{\frac{1}{2}}$,
outer radius $y^{\frac{1}{3}}$, inner radius $y^{\frac{1}{2}}$:

$$V = \pi \int_0^1 \left(y^{\frac{2}{3}} - y\right) dy = \pi \left[\frac{3y^{\frac{5}{3}}}{5} - \frac{y^2}{2}\right]_0^1$$
$$= \frac{\pi}{10}\text{u}^3$$

4

a Find the area of the region bounded by the parabola $y = x^2$ and the line $y = x + 2$.



Intersections: $x^2 = x + 2 \Rightarrow x = -1, x = 2$.

$$Area = \int_{-1}^{2} (x + 2 - x^2)\,dx = \left[\frac{x^2}{2} + 2x - \frac{x^3}{3}\right]_{-1}^{2}$$
$$= \frac{10}{3} - \left(-\frac{7}{6}\right)$$
$$= \frac{9}{2}\text{u}^2$$

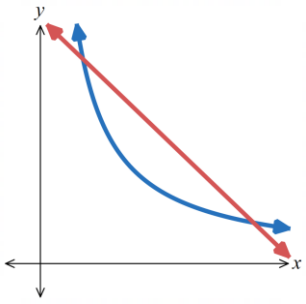
b The area bounded by the curve $y = x^2$ and the line $y = x + 2$ is rotated about the x-axis.
Find the exact volume of the solid formed.

Outer radius $= x + 2$, inner radius $= x^2$:

$$V = \pi \int_{-1}^{2} ((x+2)^2 - x^4)\,dx = \pi \int_{-1}^{2} (x^2 + 4x + 4 - x^4)\,dx$$
$$= \pi \left[\frac{x^3}{3} + 2x^2 + 4x - \frac{x^5}{5}\right]_{-1}^{2}$$
$$= \pi \left(\frac{184}{15} - \left(-\frac{32}{15}\right)\right)$$
$$= \frac{72\pi}{5}\text{u}^3$$

5

a The graphs of $xy = 5$ and $x + y = 6$ are shown. Find their points of intersection.



Intersections: $xy = 5$ and $x + y = 6$ give $x = 1, x = 5$.

b Find the volume of the solid generated when the region bounded by the two curves is rotated about the x-axis.

Outer radius $= 6 - x$, inner radius $= \frac{5}{x}$:

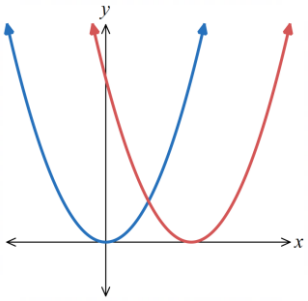
$$V = \pi \int_1^5 ((6 - x)^2 - 25x^{-2})\, dx$$

$$= \pi \left[36x - 6x^2 + \frac{x^3}{3} + \frac{25}{x}\right]_1^5$$

$$= \frac{64\pi}{3} \text{u}^3$$

6

a Find the area of the region bounded by the functions $y = x^2, y = (x - 2)^2$, and the x-axis. Curves intersect at $x = 1$; $y = x^2$ meets x-axis at $x = 0$, $y = (x - 2)^2$ meets x-axis at $x = 2$.



$$\text{Area} = \int_0^1 x^2\, dx + \int_1^2 (x - 2)^2\, dx = \frac{1}{3} + \frac{1}{3}$$

$$= \frac{2}{3} \text{u}^2$$

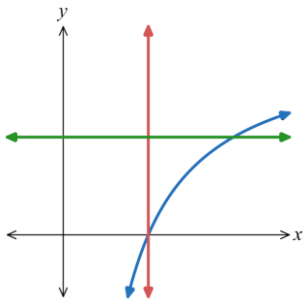
b Find the volume of the solid of revolution formed when the area enclosed between $y = x^2, y = (x - 2)^2$, and the x-axis is rotated about the x-axis.

$$V = \pi \int_0^1 x^4\, dx + \pi \int_1^2 (x - 2)^4\, dx$$

$$= \frac{\pi}{5} + \frac{\pi}{5}$$

$$= \frac{2\pi}{5} \text{u}^3$$

7 The region bounded by the hyperbola $y = 2 - \frac{2}{x}$, the vertical line $x = 1$, and the horizontal line $y = 1$ is rotated about the x-axis. Find the volume of the solid formed.



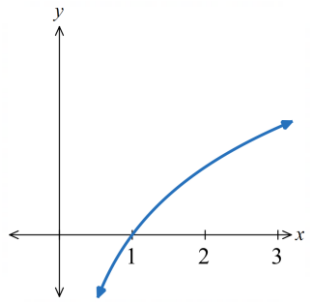
$y = 1$ when $x = 2$; outer radius = 1, inner radius $= 2 - \frac{2}{x}$, x from 1 to 2:

$$V = \pi \int_1^2 \left(1 - \left(2 - \frac{2}{x}\right)^2\right) dx = \pi \int_1^2 \left(-3 + \frac{8}{x} - 4x^{-2}\right) dx$$

$$= \pi \left[-3x + 8 \ln x + \frac{4}{x}\right]_1^2$$

$$= \pi(8 \ln 2 - 5) \text{u}^3$$

8 A rubber washer is generated by rotating the region between the curve $y = \log_e x$, the x-axis, and the line $x = 2$ about the y-axis. Find the exact volume of the washer.



$x = e^y$; outer radius = 2, inner radius = e^y, y from 0 to ln 2:

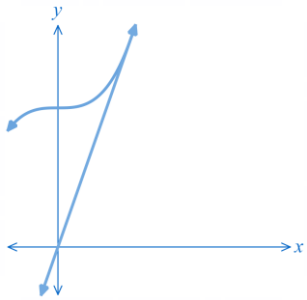
$$V = \pi\int_0^{\ln 2}(4 - e^{2y})\,dy = \pi\left[4y - \frac{e^{2y}}{2}\right]_0^{\ln 2}$$

$$= \pi\left((4\ln 2 - 2) - \left(-\frac{1}{2}\right)\right) = \frac{\pi}{2}(8\ln 2 - 3)\text{u}^3$$

9

a Find the equation of the tangent to the curve $y = x^3 + 2$ at the point where $x = 1$.

b Draw a diagram showing the region bounded by the curve, the tangent, and the y-axis.



$y' = 3x^2$; at $x = 1$: $y = 3$, gradient = 3.

Tangent: $y - 3 = 3(x - 1)$.

$y = 3x$

c Calculate the volume of the solid formed when this region is rotated about:

i the x-axis

Outer radius $x^3 + 2$, inner radius $3x$, x from 0 to 1:

$$V = \pi\int_0^1 ((x^3 + 2)^2 - 9x^2)\,dx$$

$$= \pi\int_0^1 (x^6 + 4x^3 - 9x^2 + 4)\,dx$$

$$= \pi\left[\frac{x^7}{7} + x^4 - 3x^3 + 4x\right]_0^1$$

$$= \frac{15\pi}{7}\text{u}^3$$

ii the y-axis

For y from 0 to 2:
outer radius = $\frac{y}{3}$ (tangent), inner radius = 0 (y-axis):

$$V_1 = \pi\int_0^2 \left(\frac{y}{3}\right)^2 dy = \left(\frac{\pi}{9}\right)\left[\frac{y^3}{3}\right]_0^2 = \frac{8\pi}{27}$$

For y from 2 to 3:
outer radius = $\frac{y}{3}$, inner radius = $(y - 2)^{\frac{1}{3}}$

$$V_2 = \pi\int_2^3 \left(\left(\frac{y}{3}\right)^2 - (y - 2)^{\frac{2}{3}}\right)dy$$

$$= \pi\left[\frac{y^3}{27} - \left(\frac{3}{5}\right)(y - 2)^{\frac{5}{3}}\right]_2^3$$

$$= \pi\left(\left(1 - \frac{3}{5}\right) - \left(\frac{8}{27} - 0\right)\right)$$

$$= \pi\left(\frac{2}{5} - \frac{8}{27}\right) = \frac{14\pi}{135}$$

$$V = \frac{8\pi}{27} + \frac{14\pi}{135} = \frac{2\pi}{5}\text{u}^3$$